

Deep Learning: Introduction

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Some Good Resources for DL

- Duda and Hart, "Pattern Classification" Wiley, 2001.
- C. M. Bishop, "Neural Networks for Pattern Recognition", 1995.
- B. Yegnanarayana, "Artificial Neural Networks", PHI, 1999.
- C. M. Bishop, "PRML", Springer, 2006.
- Michael Nielsen "Neural Networks and Deep Learning" Open Book.
- Ian Goodfellow, et. al. , "Deep Learning", MIT Press, 2016.
- Charu Agarwal, "NNDL", Springer 2018.
- Mithesh M. Khapra, "Deep Learning Part I and II" NPTEL Course.
- Ali Ghodsi, "Deep Learning", University of Waterloo Course.
- C. M. Bishop & H. Bishop, "Deep Learning", Springer, 2024

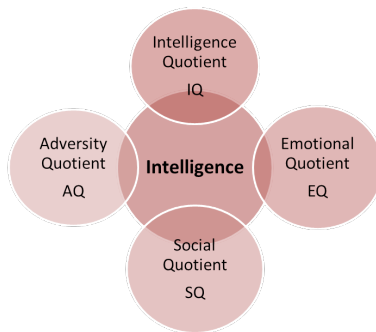


Terminologies in Machine Learning

- **Artificial Intelligence (AI):** Approaches that enable **computers** to perform **intelligent tasks**, tasks performed by humans. Ex.: Common sense, reading & comprehension, driving, learning & knowledge, remembering
- **Pattern Recognition (PR):** **Recognizing patterns** in given data. Ex: face, landmark, speech/voice, pattern vs data
- **Machine Learning (ML):** Computing approaches that **learn from data**, (**typically in the form of features**) **without being explicitly programmed on what to learn**. Pattern recognition vs machine learning
- **Deep Learning (DL):** Computing approaches that **learn from raw data**, **without being explicitly programmed on what to learn**. Machine learning vs deep learning.



- **Intelligence:** Experiencing (ability to learn & knowledge store) and use it for deciding future course. Human Intelligence vs Artificial intelligence



IQ: intellect and potential, EQ: manage emotions,

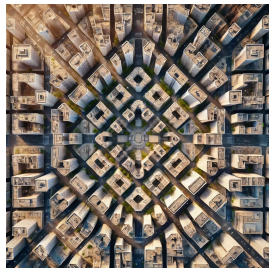
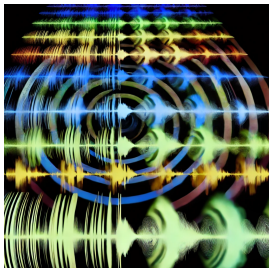
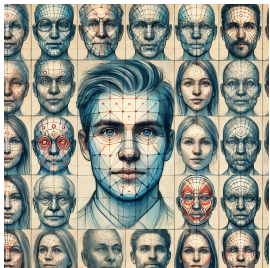
SQ: interfacing with society, AQ: ability to deal with adverse situations

- **Artificial Intelligence (AI):** Enabling machines to mimic so called intelligent tasks, performed by humans
 - Discovery, Learning, dealing with uncertainties and new situations
- **AI Categories:**
 - John Haugeland "Artificial Intelligence : The Very Idea", MIT Press, 1985
 - Symbolic AI or Classical AI - knowledge representation by symbols and rules (NPTEL AI by Deepak Khemani)
 - Connectionist AI or DL based AI - sensing, modeling & classification
 - Hybrid of Symbolic & Connectionist AI - Interpretability & scalability

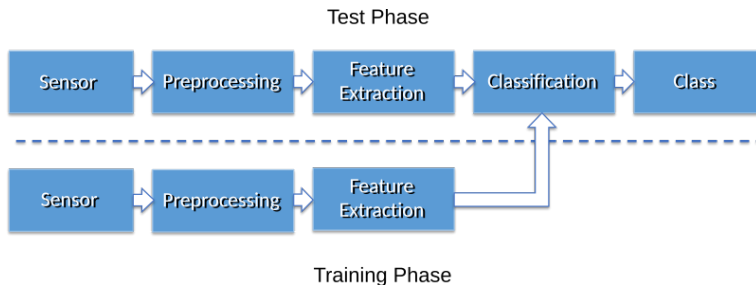
Pattern Recognition

- Task of **modeling and recognizing patterns** that are not directly obvious from raw data or features. pattern vs data, human vs computer
- Mimic human way of dealing with external world through **cognitive sensing and processing**.
- Sensing, feature selection, model creation and comparison. pattern way of processing
- Able to demonstrate pattern recognition approach.
- Template matching, Stochastic or probabilistic modeling, Neural Networks.
- Conceptually mimicked human way of dealing with natural data.
- Limited computing power and data.

Pattern Recognition Illustrations



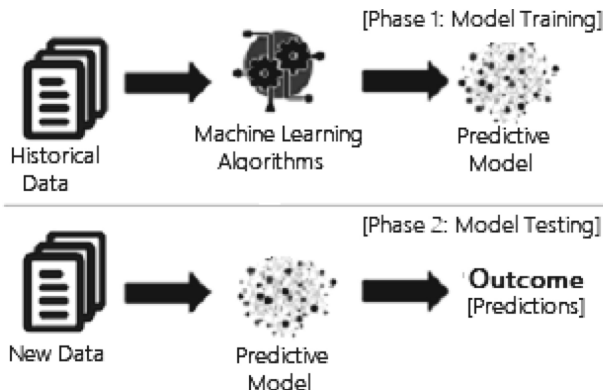
Block Diagram of Pattern Recognition System



Machine Learning

- With more and more digital data available, **task of automatic discovery and learning of information**, both natural and synthetic data
- Not much focus on feature extraction, signal processing knowledge not pre-requisite !
- More emphasis on discovery and learning of information by machine.
- Ability to discover and learn from data (features)
- Treated learning more like associated function learning.
- Output $\mathbf{y} = f(\mathbf{x})$, where $\mathbf{y}$ is output and $\mathbf{x}$ is input data (features).
- Goal of ML is to coming up with new approaches the learn $f()$ which maps $\mathbf{x}$ to $\mathbf{y}$. Linear regression, Logistic regression, Bayes classifier

Block Diagram of Machine Learning System

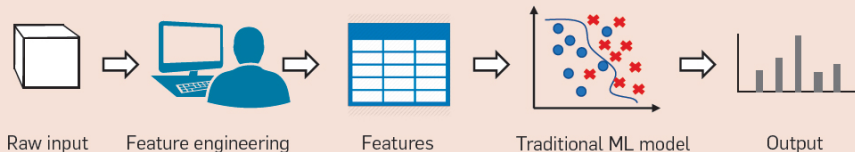


- Task of **discover and learning both features (representation) and also information** from raw data not features
- Trying to mimic human way of learning.
- Learning from experience
- Need not specify everything in the beginning
- Understand in terms of hierarchy of concepts
- Each concept defined in terms its relation to simpler concepts
- Learning complicated concepts out of simpler ones
- Learning multiple functions from same data

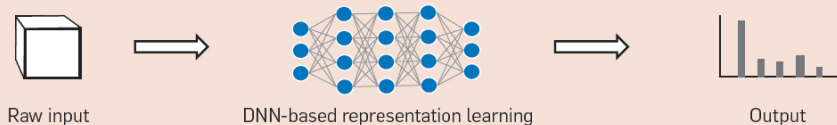


Traditional ML vs DL

Traditional machine learning



Deep learning



Pattern recognition vs machine learning vs deep learning

Neural Networks

- Inspired by the structure and functioning of human brain
- Basic computing element in human brain is Neuron
- Human cognitive power is due to billions of neurons working in a connected manner
- Parallel and distributed processing
- Mimicked same using simple artificial neuron and neural network architectures
- Performance is not good even to compete with contemporary stochastic models
- Hence the motivation for deep learning

Neural Nets vs Deep Learning

- The concept of deep learning first originated from neural network.
- May be as advanced neural learning concept
- A good example of deep neural network is a deep feed forward neural network (DFFNN or DNN).
- Backpropagation (BP) is the workhorse algorithm for learning the parameters of FFNN.
- **BP did not work well for networks** having more than a small number of hidden layers.
- **Insufficient data leading to overfitting and difficulty in training** of the deep networks was the main limitation.



History and Breakthroughs of Deep Learning

- **Geoffrey Hinton** demonstrated that a deep belief network could be efficiently trained using a greedy layer-wise pre-training (2006)
- Other research groups, reinforced in other deep networks (2007)
- Able to train deeper neural networks, not possible before
- DNNs outperformed systems based on other machine learning technologies & hand-crafted features
- **Learning strategies and models** have undergone changes that simplify the training of very deep architectures.



Neural Networks and Deep Learning

- Genesis of ML is in PR.
- Genesis of DL is in NN.
- Like PR is basis for ML.
- NN is basis for DL.
- NN models information for pattern recognition from features.
- DL enables both learning features and also modeling information.
- Depth is not only in terms of no of layers, but also in terms of **no of learning functions (concepts) and relating different concepts.**
- $DL \in ML \in AI$.

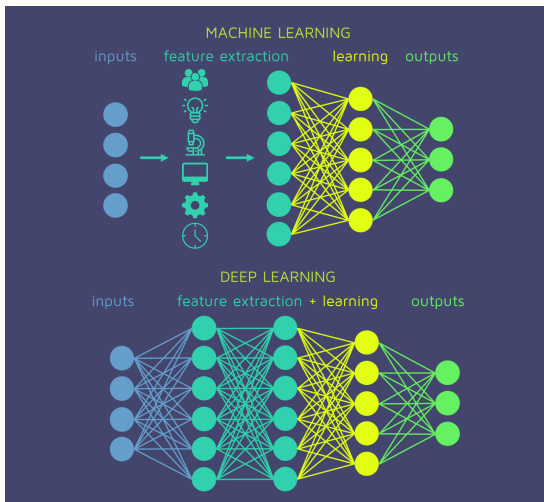


Why Deep Learning did not work earlier?

- Unable to train
- Insufficient training data
- Limited computing power
- Turn of 21st century changed the scenario
- Breakthrough in terms of training DL model
- Easy to collect data in connected world
- Large computing power at disposal due to GPU



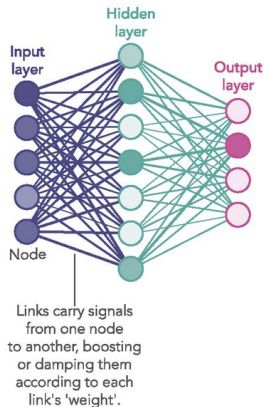
ML vs DL



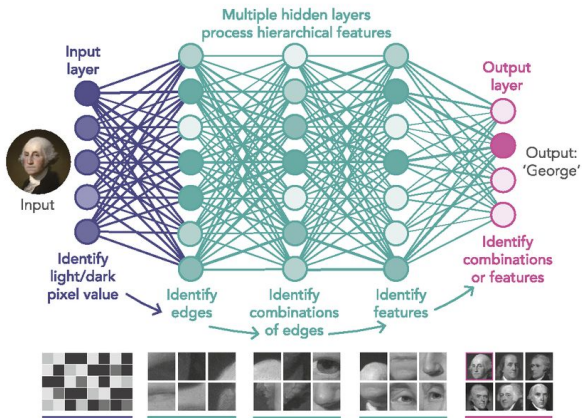
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NN vs DL

1980S-ERA NEURAL NETWORK

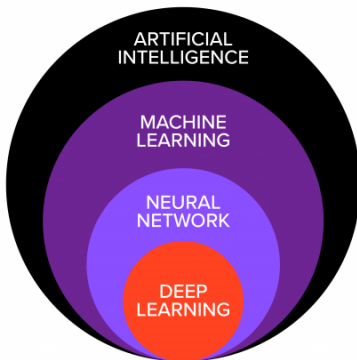


DEEP LEARNING NEURAL NETWORK



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NEURAL NETWORKS AND ARTIFICIAL INTELLIGENCE



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Applications of Deep Learning

- Deep learning uses multilayered neural networks trained with large data sets to solve complex information processing tasks
- Revolutionized many domains including computer vision, speech recognition, and natural language processing
- Growing applications across healthcare, manufacturing, commerce, finance, scientific discovery, and many other sectors.
- Massive neural networks, i.e., large language models, having order of a trillion learnable parameters, found to exhibit first indications of general artificial intelligence
- LLMs drive biggest disruptions in the history of technology.
- GPT (Chat-GPT) seems to be the biggest disrupter in technology

